SHIVAM KAMBOJ

Quantum Theorist · Open Systems & QEC Architecture

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First-Principles Quantum Theorist & Computational Physicist (Ph.D. Expected Fall 2026): Modality-agnostic theorist working from microscopic Hamiltonians up to fault-tolerant logical architectures. Combines symmetry-resolved sparse exact diagonalization on HPC clusters, Lindblad and non-Markovian open-system dynamics, and production surface-code QEC software (**Stim, PyMatching**) with hands-on LLNL hardware process tomography.

CORE TECHNICAL COMPETENCIES

Neutral Atom · Trapped Ion · Photonic · Superconducting

- **Quantum Theory & Error Correction:** Surface-code QEC benchmarking, parity-protected bosonic & cat qubits, non-Markovian & temporally correlated $1/f$ noise, Lindblad master equations, ultrastrong light-matter coupling, Bogoliubov & normal-mode analysis.
- **Numerical Solvers & QCVV:** Symmetry-resolved sparse Exact Diagonalization (CSR/Krylov solvers), Fock-space truncation convergence, two-layer Monte Carlo threshold sweeps, bootstrap confidence intervals, Linear-Inversion Quantum Process Tomography (LI-QPT).
- **Production Software & HPC Stack:** Python, Stim, PyMatching, QuTiP, SciPy, NumPy, Qiskit, SLURM (HPC), Git, CI/CD, `mypy --strict`, `ruff`, `pytest`, Flask, Google Gemini API, GCP, COMSOL RF & AC/DC.

FAULT-TOLERANT QEC & PRODUCTION SCIENTIFIC SOFTWARE

PyPI Release · Strict CI/CD

`nonmarkov_qec` — Open-Source Surface-Code QEC Benchmarking Library (Released on PyPI) 2026 – Present

- **End-to-End QEC Pipeline:** Architected a modular Python 3.11+ package integrating Ornstein-Uhlenbeck and sum-of-OU $1/f$ noise trajectories, compiled-template **Stim** noise injection (35× speedup), **PyMatching** MWPM decoding, and two-layer Monte Carlo threshold sweeps with parametric-bootstrap 95% confidence intervals.
- **Correlated Hardware Noise Analysis:** Benchmarked temporally correlated dephasing against statistically matched white-noise baselines across rotated surface codes ($d = 3, 5$) to isolate temporal correlation effects from marginal error-strength changes.
- **Software Engineering Rigor:** Enforced 74 unit and analytic-validation tests (`pytest`), strict static typing (`mypy --strict`), linting (`ruff`), and automated GitHub Actions CI/CD releases.

Physics Concept Analyzer Tool (PCAT) — Cloud AI Reasoning Platform (Python, Flask, Gemini API, GCP) 2025

- Designed and deployed a full-stack web application on Google App Engine automating structured first-principles physical model construction and assumption verification.

RESEARCH & NATIONAL LABORATORY EXPERIENCE

First-Principles Theory to Hardware Validation

Tian Group, University of California, Merced — *Graduate Student Researcher (Advisor: Prof. Lin Tian)* **May 2023 – Present**

- **Bosonic Cat Qubits & Symmetry Protection:** Combine analytical normal-mode theory and symmetry-resolved sparse exact diagonalization to resolve low-energy spectra by parity and site-exchange symmetry, distinguishing level crossings and stabilizing a parity-protected bosonic cat-state doublet.
- **Open-System Master Equations at Scale:** Simulate state evolution, macroscopic entanglement, and dissipation in QuTiP; enforce Hilbert-space truncation convergence and execute checkpointed parameter sweeps on SLURM HPC clusters.
- **Device-Level Parameterization:** Build COMSOL electromagnetic models to extract capacitances, eigenmode frequencies, impedances, and inductive participation ratios to bound realistic circuit Hamiltonians.

Lawrence Livermore National Laboratory (LLNL) — *Physics Graduate Intern, Quantum Coherent Device Group* **May – Aug 2024**

- **Quantum Process Tomography & Pulse Control:** Developed Python software for pulse-level control of superconducting qudits and engineered a generalized Linear-Inversion QPT pipeline using pseudo-inverse reconstruction to map noisy experimental data into process matrices.
- **Automated Calibration:** Automated Ramsey and T_1/T_2 fitting routines to provide rapid quantitative feedback for control experiments.

Sharping Group (UC Merced) & Hurst Group (SJSU / UC Merced) — *Collaborator & Graduate Researcher* **Jun 2021 – Present**

- Formulated Hamiltonian models of coupled YIG-cavity magnon-photon systems with parametric driving (Sharping Group) and derived analytical dispersion relations and numerical solvers for disorder-robust edge modes in 2D spin-

torque arrays (Hurst Group, published in *Phys. Rev. B*).

EDUCATION & SELECTED PUBLICATIONS

Available for Fall 2026 Industry Transition

Ph.D. in Physics (Quantum Computing), UC Merced (Expected Fall 2026) · **M.S. in Physics**, San Jose State University (2022)

1. S. Kamboj, R. A. Duine, B. Flebus, and H. M. Hurst, "Oscillatory edge modes in two-dimensional spin-torque oscillator arrays," *Phys. Rev. B* **109**, 094436 (2024).

2. S. Kamboj and L. Tian, "Deterministic Cat States and Parity-Protected Bosonic Qubits in the Quantum Rabi Dimer" (In prep, 2026).

3. S. Kamboj, J. Beurkens, J. Sharping, and L. Tian, "Superconducting realization of the Quantum Rabi Dimer" (In prep, 2026).